

13 Percentiles: Cumulative Distribution Functions and Quantile Functions

- The probability density function of a continuous random variable describes which values are more or less likely in the “close to” sense.
- However, the pdf does not provide probabilities directly; rather, it is areas under the pdf which provide probabilities.
- There are many ways of describing distributions. In particular, a distribution is basically just a collection of percentiles.
- Cumulative distribution functions and quantile functions describe distributions by specifying all the percentiles.
- Roughly,
 - The *cumulative distribution function (cdf)* of a random variable fills in the blank for any given x : x is the (blank) percentile.
 - The *quantile function* fills in the following blank for a given $p \in (0, 1)$: the $(100p)$ th percentile is (blank).

Example 13.1. Explain what the percentiles mean in the following statements.

- 47cm is 12th percentile of heights for newborn girls.
 - 620 is the 80th percentile of SAT Math scores.
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- A distribution is characterized by its percentiles.
 - Roughly, the value x is the p -th **percentile** (a.k.a. quantile) of a distribution of a random variable X if p percent of values of the variable are less than or equal to x : $P(X \leq x) = p$.
 - A few commonly used percentiles
 - First (or lower) quartile is the 25th percentile ($p = 0.25$)
 - Median is the 50th percentile ($p = 0.50$)

- Third (or upper) quartile is the 75th percentile ($p = 0.75$)

Example 13.2. Recall Example 12.1 and Example 12.2 where office arrival times, measured in minutes after 8am, followed this pdf

$$f_X(x) = \begin{cases} (2/3600)x, & 0 < x < 60, \\ 0 & \text{otherwise.} \end{cases}$$

1. Compute $P(X < 15)$ and phrase the result in terms of a percentile.
2. Compute $P(X < 45)$ and phrase the result in terms of a percentile.
3. Find an expression for $P(X \leq x)$ for general x . This function, denoted $F_X(x)$, is called the *cumulative distribution function (cdf)* of X .
4. How does the cdf relate to percentiles?
5. Use the cdf to determine what percentile 30 minutes after 8am is.
6. Compute and interpret $F(40) - F(30)$.

- The *cumulative distribution function (cdf)* of a random variable fills in the blank for any given x : x is the (blank) percentile. That is, for an input x , the cdf outputs $P(X \leq x)$.
- The **cumulative distribution function (cdf)** of a random variable X is the function F_X defined by $F_X(x) = P(X \leq x)$. A cdf is defined for all real numbers x regardless of whether x is a possible value of X .

Example 13.3. Continuing Example 13.2 where X (minutes after 8am) has pdf

$$f_X(x) = (2/3600)x, \quad 0 < x < 60.$$

and cdf

$$F_X(x) = \begin{cases} 0, & x < 0, \\ (x/60)^2, & 0 < x < 60, \\ 1, & x > 60. \end{cases}$$

1. Find and interpret the 10th percentile.
2. Find and interpret the 90th percentile.
3. Find an equation for the $(100p)$ th percentile (e.g., $p = 0.9$ for the 90th percentile).

- The *quantile function* (essentially the inverse cdf) fills in the following blank for a given $p \in (0, 1)$: the 100 p th percentile is (blank).
- For example, evaluating the quantile function at $p = 0.25$ outputs the 25th percentile.
- For a *continuous* random variable X with cdf F_X , the **quantile function** Q_X is the inverse of the cdf, $Q_X(p) = F_X^{-1}(p)$.
- A quantile function basically gives you the same information as a cdf, just in the other direction.

- Both a quantile function and a cdf define a distribution by specifying all the percentiles.

Example 13.4. Continuing Example 13.3 where X (minutes after 8am) has pdf

$$f_X(x) = (2/3600)x, \quad 0 < x < 60.$$

and cdf

$$F_X(x) = \begin{cases} 0, & x < 0, \\ (x/60)^2, & 0 < x < 60, \\ 1, & x > 60. \end{cases}$$

and quantile function

$$Q(p) = 60\sqrt{p}, \quad 0 < p < 1.$$

1. Construct a circular spinner for simulating values of X . Version 1: label the values 0, 10, 20, 30, 40, 50, 60. Careful: they will not be equally spaced.
2. Sketch a histogram with equal bin widths—0 to 10, 10 to 20, etc.—that you would expect to see after spinning the version 1 spinner many times. Does it have the proper shape?
3. Construct a circular spinner for simulating values of X . Version 2: label the circular axis so that your spinner has 12 sectors each with probability $1/12$.
4. Sketch a histogram that you would expect to see after spinning the version 2 spinner many times. Start by sketching a histogram with *unequal* bin widths, then sketch the correspond equal bin width histogram. Does it have the proper shape?

- The cdf and the quantile function can be used to create a spinner for a distribution.
- A spinner basically describes a distribution by specifying all the percentiles. For example,
 - the 25th percentile goes 25% of the way around (“3 o clock”)
 - the 50th percentile goes 50% of the way around (“6 o clock”)
 - the 75th percentile goes 75% of the way around (“9 o clock”)
- Intervals corresponding to regions of higher density (“more likely” values) are stretched out on the spinner boundary; intervals corresponding regions of lower density (“less likely” values) are shrunk.
- **Universality of the Uniform (or “one spinner to rule them all”).** Let F be a cdf and Q its corresponding quantile function. Let U have a Uniform(0, 1) distribution and define the random variable $X = Q(U)$. Then the cdf of X is F .
- Universality of the uniform might look complicated but all it basically says is that you can construct a spinner by putting the 25th percentile 25% of the way around, the 75th percentile 75% of the way around, etc.
- Actually, universality of the uniform says we don’t have to create a new spinner. We can just spin the Uniform(0, 1) spinner, with evenly spaced values from 0 to 1, and transform each resulting value by plugging it into the quantile function.

Example 13.5. Continuing Example 12.3. Suppose that X is a claim amount in excess of the deductible for a car insurance policy that incurs a claim, measured in thousands of dollars. Assume that X has pdf

$$f_X(x) = (1/4.3)e^{-x/4.3}, \quad x > 0$$

1. Compute $P(X < 5)$ and phrase the result in terms of a percentile.
2. Find an expression for the CDF of X .
3. Compute the 25th percentile.

4. Find an expression for the quantile function of X .

5. Use the quantile function to find the 75th percentile.

13.1 Exercises

Exercise 13.1. Continuing Exercise [12.1](#). In a certain population, household income X (\$ thousands) follows the pdf

$$f_X(x) = cx^{-2.5}, \quad x \geq 30$$

for an appropriate constant c . Note that the measurement units are \$ thousands, so 1 represents 1 thousand dollars. (This is not a super realistic example, but just go with it.)

1. Compute $P(X \leq 100)$ and phrase the result in terms of a percentile.

2. Find the cdf of X .

3. Find the 10th percentile.

4. Find the quantile function of X .

5. Sketch a circular spinner for simulating values of X .

Exercise 13.2. Let X be the number of heads in 3 flips of a fair coin.

1. Find the pmf of X and sketch a plot of it.
 2. Find the cdf of X and sketch a plot of it.
 3. Let Y be the number of tails in 3 flips of a fair coin. Find the cdf of Y .
- A cdf is a non-decreasing function: if $x_1 \leq x_2$ then $F_X(x_1) \leq F_X(x_2)$.
 - A cdf approaches 0 as the input approaches $-\infty$: $\lim_{x \rightarrow -\infty} F_X(x) = 0$
 - A cdf approaches 1 as the input approaches ∞ : $\lim_{x \rightarrow \infty} F_X(x) = 1$
 - The cdf of a *discrete* random variable is a step function.
 - The steps occur at the possible values of the random variable.
 - The height of a particular step corresponds to the probability of that value, given by the pmf.
 - The cdf of a *continuous* random variable is a continuous function.
 - The cdf of a *continuous* random variable is obtained by integrating the pdf, so

- The pdf of a *continuous* random variable is obtained by differentiating the cdf

$$F'_X = f_X \quad \text{if } X \text{ is continuous}$$

- For any random variable X with cdf F_X

$$F_X(b) - F_X(a) = P(a < X \leq b)$$

Whether the inequalities in the above event are strict ($<$) or not ($\leq$) matters for discrete random variables, but not for continuous.

- Random variables X and Y **have the same distribution** if their cdfs are the same, that is, if $F_X(u) = F_Y(u)$ for all u .
- That is, two random variables have the same distribution if all the percentiles are the same.